

M&E SCHEMA AND IMPORT GUIDE

This document explains what the current M&E module stores, how weekly Excel import works, and what each major table does.

1) CORE DATA MODEL

ME_PROJECTS

- ? Purpose: M&E project master list.
- ? Key fields:
- ? name
- ? project_code (unique project reference)
- ? description, start_date, end_date

ME_REPORTING_PERIODS

- ? Purpose: reporting windows used by weekly/baseline/midline/endline entries.
- ? Key fields:
- ? project_id
- ? type (weekly, baseline, midline, endline)
- ? start_date, end_date, label
- ? is_locked

ME_INDICATORS

- ? Purpose: indicator definitions.
- ? Key fields:
- ? project_id (nullable)
- ? code (unique)
- ? name, name_local
- ? framework_type (output, outcome, impact)
- ? unit, frequency
- ? direction, data_type
- ? alert thresholds (threshold_warning, threshold_critical)

ME_INDICATOR_TARGETS

- ? Purpose: target values by indicator and period/date scope.
- ? Key fields:
- ? indicator_id
- ? reporting_period_id (nullable)
- ? period_start, period_end
- ? target_value, target_min, target_max
- ? scope_location, scope_project

ME_INDICATOR_REPORTS

? Purpose: actual values by indicator and period/date scope.

? Key fields:

? indicator_id

? reporting_period_id (nullable)

? period_start, period_end

? actual_value (+ actual_text optional)

? scope_location, scope_project

? source (manual, import, survey, api)

? entered_by, entered_at, comment, notes

2) DISAGGREGATION

ME_DISAGGREGATION_CATEGORIES

? Dimension types (gender, age, location, disability, custom).

ME_DISAGGREGATION_OPTIONS

? Allowed options under each category.

ME_INDICATOR_DISAGGREGATION

? Which categories are enabled for each indicator.

ME_REPORT_DISAGGREGATION_VALUES

? Actual values per report/category/option.

ME_INDICATOR_TARGET_DISAGGREGATIONS

? Target values per target/option.

3) ALERTS

ME_ALERT_RULES

? Configurable rule thresholds.

? Supports both old condition style and new rule metadata:

? condition, warning_threshold, critical_threshold

? rule_type, threshold, severity, is_active

ME_ALERTS

? Generated alerts for report performance.

? Key fields:

? project_id, indicator_id, report_id, reporting_period_id

? severity, status, message

? resolved_at, resolved_by

4) SURVEYS

ME_SURVEYS

? Survey definition (weekly/baseline/midline/endline).

? Linked optionally to project_id and reporting_period_id.

ME_SURVEY_QUESTIONS

? Survey questions and optional mapping to indicators (maps_to_indicator_id).

ME_SURVEY_RESPONSES

? Individual submissions with source and submitter/project/period linkage.

ME_SURVEY_ANSWERS

? Answers per response/question, including answer_value JSON.

5) AUDIT AND FEEDBACK

ME_AUDIT_LOGS

? Change history for M&E records.

? Stores action + payload (changes, and optional before/after).

ME_BENEFICIARY_FEEDBACK

? Feedback entries with optional project/period/location and disaggregation option refs.

ME_LOCATIONS

? Optional location hierarchy (parent_id) with map coordinates (lat, lng).

6) WEEKLY EXCEL IMPORT (CURRENT BEHAVIOR)

Import page: M&E -> Import Weekly Report

Service: App\Services\ME\WeeklyReportImportService

ACCEPTED FILE SHAPE

? Wide Google-Form-like weekly sheet (Amharic headers supported).

- ? Header row contains:
- ? timestamp
- ? project name
- ? project code
- ? reporting period text
- ? many paired metric columns (plan/actual)

WHAT IMPORTER DOES

1. Reads sheet 1 and detects core columns.
2. Detects metric columns by plan/actual keywords.
3. Normalizes project codes and period ranges.
4. Creates missing me_projects and me_reporting_periods (when table exists).
5. Auto-creates missing me_indicators for detected metrics.
6. Upserts:
 - ? plan values -> me_indicator_targets
 - ? actual values -> me_indicator_reports (source=import)
7. Writes debug JSON report to:
 - ? storage/app/me-import-reports/weekly-import-YYYYMMDD_HHMMSS.json

IDEMPOTENCY

- ? Re-importing same file updates existing records (upsert) rather than duplicating rows.

7) OBSERVER BEHAVIOR

Observer: App\Observers\ME\MeIndicatorReportObserver

- ? On report save, it attempts to synchronize alerts through AlertService.
- ? It safely skips/ignores sync when alert tables are unavailable during setup/import bootstrap.

8) NOTES FOR FUTURE CHANGES

- ? If headers are renamed later (for example fully English), importer still works as long as:
 - ? project name/code and period columns are present
 - ? metric columns still include recognizable plan/actual semantics
- ? For strict template enforcement, implement a dedicated header map and validation mode.